

I want to keep on exercising but I don't: The negative impact of momentary lacks of self-control on exercise adherence

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ABSTRACT

Objectives: We investigated the impact of state and trait self-control strength on exercise execution. We hypothesized that state self-control strength is reduced on days that are perceived as stressful and that reductions in self-control strength result in a lower likelihood to work out. Additionally, trait self-control strength may affect the execution of workout plans.

Design: University students, who stated that they have not been exercising regularly for at least one month, filled in a trait self-control inventory, a personality questionnaire, and were instructed to perform a daily workout over a one-week period. Perceived stress levels, state self-control strength, and workout completion were assessed on a daily basis.

Results: Results revealed that people were less likely to exercise on days they perceived as stressful. State self-control mediated the relation between stress and exercise completion. Trait self-control and other personality variables did not affect workout completion.

Conclusions: Results indicate that daily stress is associated with self-control depletion and a lower likelihood to work out.

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1. Introduction

Regular physical activity can be beneficial for one's health and subjective well-being (e.g., Biddle & Mutrie, 2007; Ströhle, 2009) and is associated with better mental health, lower levels of stress, and lower levels of anxiety (e.g., Bhui & Fletcher, 2000; Goodwin, 2003). Most medical rehabilitation programs include some sort of physical activity (e.g., Dunn, Trivedi, & O'Neal, 2001). On the contrary, a sedentary lifestyle is associated with negative consequences, such as overweight, obesity, type-2 diabetes mellitus, and cardiovascular diseases (e.g., Hamilton, Healy, Dunstan, Zderic, & Owen, 2008). The World Health Organization's (WHO) criteria for a healthy lifestyle include at least 150 min of moderate-intensity aerobic physical activity per week. Adults are recommended to keep their activity levels to 300 min per week for additional health benefits (WHO, 2010). However, a recent study revealed that 31% of adults worldwide are physically inactive (i.e., less than 150 min of physical activity) and that 80% of the 13–15-year-olds engage in

physical activities for less than 60 min per day (Hallal et al., 2012). Furthermore, maintaining an active lifestyle over time seems to be difficult as daily hassles (i.e., stressful encounters during the day) can often interfere with the execution of workout plans. Nguyen-Michel and colleagues have shown, for instance, that higher levels of stress are associated with lower levels of physical activity (Nguyen-Michel, Unger, Hamilton, & Spruijt-Metz, 2006).

Although most individuals are aware of the positive effects of regular physical activity and have the intention to work out regularly, several studies demonstrate that there is often a gap between workout intentions and behaviour—that is, workout behaviour cannot be maintained in the longer term (e.g., Rhodes, Plotnikoff, & Courneya, 2008). Most existing traditional theoretical approaches of physical activity (e.g., Theory of Planned Behaviour; Ajzen, 1991) cannot fully explain lapses in exercise behaviour, as they primarily focus on how exercise intentions are formed (e.g., Sniehotta, Presseau, & Araújo-Soares, 2014; see also; Rebar, Loftus, & Hagger, 2015). However, especially with regard to workout plans, it is often the case that despite one's best intention, plans are not executed (Rhodes et al., 2008). This highlights the need of identifying predictors of exercise execution. Martin Ginis and Bray (2010) propose that the ability to adhere to an exercise schedule over an

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extended period of time is an act of self-control (see also Bandura, 2005). In line with this notion, more recent models of physical activity also include post-intentional self-regulatory acts of shielding intentions from distracting or tempting alternatives (e.g., relaxation instead of working out after a long day at work; cf. Gollwitzer, 1999; Schwarzer, 1992; Schwarzer & Luszczynska, 2008). In the present research, we build on a further recent theoretical approach to explain lapses in exercise behaviour: the *Strength Model of Self-Control* (Baumeister, Bratslavsky, Muraven, & Tice, 1998; see also Hagger, 2010; Hagger, Wood, Stiff, & Chatzisarantis, 2009; Martin Ginis & Bray, 2010).

1.1. The Strength Model of Self-Control

Baumeister and colleagues define self-control as the ability to inhibit or alter predominant response tendencies and to resist immediate gratifications in favour of more desirable long-term benefits (e.g., Baumeister et al., 1998). Accordingly, any self-control act (e.g., emotion regulation, attention regulation) is energized by one global metaphorical resource (e.g., Schmeichel & Baumeister, 2010). This self-control resource, however, is limited and people differ regarding their self-control capacities (i.e., trait-self-control strength; Tangney, Baumeister, & Boone, 2004). Additionally, self-control strength can also become temporarily depleted; for example, after a straining self-control act was performed (i.e., *ego depletion*; e.g., Baumeister et al., 1998). Ego depletion is assumed to negatively affect subsequent behaviour that requires self-control. This idea is supported by a recent meta-analysis showing medium to large sized ego-depletion effects (Hagger, Wood, Stiff, & Chatzisarantis, 2010).

Recent studies showed that state self-control strength may also be of central significance in the sport and exercise domain (cf. Englert & Bertrams, 2012). For instance, Dorris, Power, and Kenefick (2012) found athletes to perform fewer push-ups and fewer sit-ups in a state of ego depletion as compared to a state with fully available self-control strength. In light of such findings, it becomes apparent that states of ego depletion render it unlikely that individuals push themselves to the limit, although it would be physiologically possible. This general finding may not only be of relevance for athletes but also for individuals intending to maintain a healthy active life style. For these individuals, working out may require self-control and thus draws on limited self-control resources as one needs to inhibit predominant tendencies to relax and to instead engage in other—potentially exhausting—actions: **Importantly, many individuals first have to deal with a number of different stressful events during a day, before they can even initiate their workout plans as part of their leisure activities. It has been argued that higher levels of perceived stress are associated with a higher likelihood of depleted self-control strength** (cf., Tangney et al., 2004). Such a state of low self-control strength may, in turn, render a workout execution less likely (cf., Ntoumanis, 2014).

A recent study by Bertrams and Englert (2013) showing that participants with lower levels of trait self-control strength were less likely to meet their physical-activity standards provides first empirical support for this assumption. Additionally, Stork and colleagues found out that self-reported as well as objectively measured trait self-control strength significantly predicted exercise behaviour over a 4-week period (Stork, Graham, Bray, & Martin Ginis, 2016). Based on these findings, we assume that individuals with lower trait self-control strength will be less likely to be physically active. However, self-control strength does not only vary between individuals but also within individuals between situations (e.g., Tangney et al., 2004). That is, there is a situational (i.e., state) and a dispositional (i.e., trait) component of self-control strength and both components may explain parts of the variance in the

execution of workout intentions (cf., Baumeister et al., 1998; Tangney et al., 2004). As the former component lies more under personal control, it is important to know whether it affects the execution of workout acts that may come with health benefits in the long run, but are exhausting in the short run. **A study by Martin Ginis and Bray (2010) provides first indications that state self-control strength plays a role in the execution of workout plans. In this study, the authors found that prior self-control exertion lead to decreases in the intended effort to engage in future exercise, and was associated with a lower likelihood of adhering to exercise plans.** Therefore, we were interested in whether state self-control strength would predict whether an individual executes his/her workout schedule on a given day or not. Furthermore, we investigated whether trait self-control strength explains variance in workout execution over-and-above state self-control strength and whether inter-individual differences in trait self-control strength influence the relationship between state self-control strength and workout execution (for a discussion on the role of trait and state self-control strength, see also Hofmann, Baumeister, Förster, & Vohs, 2012).

Further, to gain a better understanding of why self-control is sometimes situationally reduced, we also assessed how stressed individuals perceived their day. As dealing with stressful events requires self-control, state self-control strength should be reduced after stressful episodes (e.g., Tangney et al., 2004). Accordingly, we predicted a mediation model that stressful events should temporarily deplete state self-control strength and that this depletion should, in turn, render workout execution less likely (see Fig. 1, for an illustration). Finally, we also considered inter-individual differences in social desirability, self-efficacy, and global personality dimensions as potential predictors, because these personality variables have been shown to influence workout behaviour in other studies (e.g., Bandura, 2005; Conner, Rodgers, & Murrey, 2007).

2. Method

2.1. Participants

According to the WHO (2010; see also above) at least 150 min of moderate-intensity aerobic physical activity per week are recommended for a healthy lifestyle, while 300 min per week are considered beneficial for additional health benefits. We therefore aimed to only include participants in our final sample whose self-reported physical activity was currently below 300 min per week and who should thus gain additional health benefits from additional exercising. Furthermore, we only included participants that have not been exercising regularly for the last month and who explicitly stated that they wanted to be more physically active. To identify the individuals of interest, potential participants were first invited to take part in an activity-level screening via mailing lists and adverts. A total of 449 participants were screened; most of them were students of or worked at Heidelberg University. Of this sample, 107 persons matched the criteria outlined above and were thus invited to participate in the study. From this sample, 62 individuals agreed to further participate in the present study. Demographic information and information regarding the activity status of the sample is provided in Table 1.

Participants received a monetary compensation of up to 40 Euros depending on their compliance. The study was carried out in accordance with the Helsinki Declaration of 1975 and we collected written informed consent from each participant.

2.2. Procedure

Individuals who were physically active for less than 300 min per

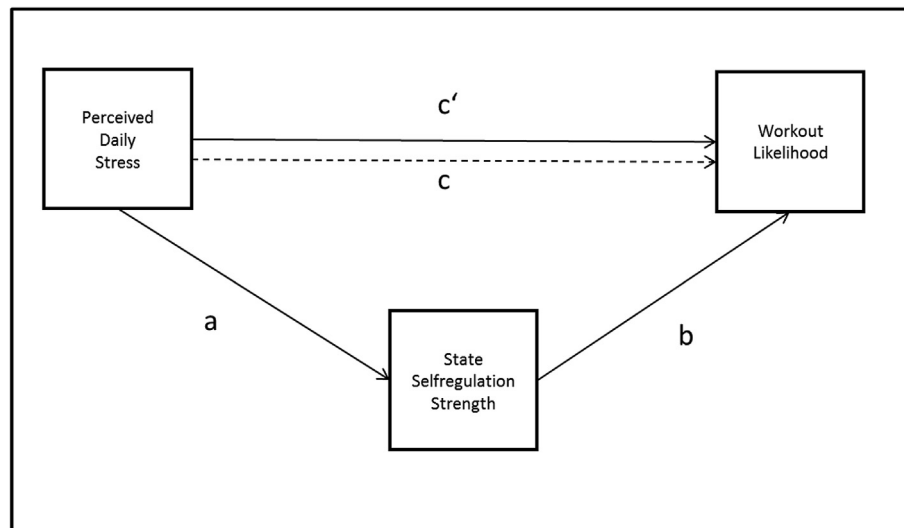


Fig. 1. Illustration of the mediational pathway from perceived daily stress via state self-control strength to workout likelihood.

Table 1
Demographic information and information on activity levels of the present sample.

Measures	<i>M</i>	<i>SD</i>	Range
Age (years)	23	4	18–34
Gender (% female)	79	—	—
Body height (cm)	171	9	155–196
Body weight (kg)	65	13	42–117
Body mass index (kg/m ²)	22	3	17–35
Physical activity (non-sport; min/week)	134	69	15–270
Physical activity (sports; min/week)	0	—	—

Note. Physical activity refers to the last four weeks. Non-sport activities included daily activities like walking, biking, and housework. None of the selected participants engaged in any sports activities (i.e., any type of regular exercise that participants considered themselves as sport activities).

week, who did not engage in regular exercising, and who wanted to be more physically active were invited to the laboratory. Participants were scheduled in sessions of one or two. In the pre-study laboratory session, participants first generated a unique personal code that was used to combine the data from the different sessions and time-points and filled in a basic demographic questionnaire. Next, we assessed participants' health status, trait self-control strength, while we also applied a measure of social desirability and a memory questionnaire that was not of further interest for the present study. The experimenter then evaluated whether participants were physically able to perform the workout. Evaluation of all participants was positive. Participants next received a workout manual and practiced all exercises with an instructor. The workout lasted approximately 15 min. Participants were asked to execute two sets of each exercise as a daily workout between 6 and 10 p.m. for one week, starting the next day and all participants agreed to do so. Participants were further asked to fill in a daily online questionnaire for which they received a link. Participants received 10 Euros for their participation in the pre-study session. They were informed that they would receive another 20 Euros for their participation in the study and an additional 10 Euros if they did not skip more than one of the daily questionnaires. It was ensured that all participants were aware that the monetary incentives were not tied to the workout execution but to questionnaire completion.

For the following seven days, participants had to fill in daily online questionnaires asking them to first provide their personal code and then to indicate for each exercise whether they performed

two sets, one set, or did not perform it at all. Furthermore, participants' state self-control strength and their stress-levels experienced during the course of the day were assessed as part of the daily questionnaires, assuming that higher levels of perceived stress during the day would be associated with lower levels of state self-control strength in the evening. Routinely, participants received a reminder to not forget last day's questionnaire on the following day via email and where allowed to fill it in before noon. On the seventh day, we additionally assessed self-efficacy and several personality traits in addition to the regular daily questions. Finally, participants were debriefed and received their payment.

2.3. Measures

Physical activity screening. Pre-study activity levels were assessed with a standardized questionnaire (Bewegungs-und Sportaktivität Fragebogen; BSA; Fuchs, Klaperski, Gerber, & Seelig, 2015). Participants were instructed to indicate for how many minutes per week they performed everyday physical activities (e.g., taking the bike to work; taking a walk) in average during the last four weeks. Participants who reported being physically active for less than 300 min per week, have not been working out regularly for at least one month, and were motivated to increase their activity level were invited to take part in the study.

Health status. The current health status was assessed via the German version of the Physical Activity Readiness Questionnaire (PARQ; Thomas, Reading, & Shephard, 1992), which consists of seven dichotomous items (e.g., 'Have you experienced any sort of chest pain during the last month?'; yes vs. no). Individual response patterns were used for a guided interview to assess whether participants were physically able to perform the workout.

Social desirability scale. To control for potential social-desirability effects on self-reports we applied a social desirability scale (SES-17; Stöber, 1999). Participants answered 17 dichotomous items (e.g., 'I always admit my mistakes openly and face the potential negative consequences'; yes vs. no). Individual sum scores across all items were used as a social-desirability trait measure.

Trait self-control strength. Trait self-control strength was assessed with the German short version of the Self-Control Scale (SCS-K-D; Bertrams & Dickhäuser, 2009). The scale consists of 13 items (e.g., 'I am good at resisting temptations') that have to be answered using 5-point Likert-type scales (1 - not at all to 5 - very

much). The mean response to all items was used as a measure of trait-self-control strength.

Workout manual and diary. The workout manual contained seven different standard bodyweight exercises that did not require any additional sport equipment. These exercises were chosen based on the recommendations of two certified athletic trainers in order to be challenging but feasible for individuals that do not work out regularly. Every exercise was verbally described and picture illustrated (cf. Fig. 2; see also Table S2 in the Supplemental Material, for all exercises). In the online workout diary, participants were prompted with all pictures from the workout manual and had to indicate for each exercise whether they performed two sets, one set, or did not perform this exercise at all. The sum of all sets performed was used to assess workout completion.

State self-control strength. State self-control strength was assessed with the German short version of the State Self-Control Capacity Scale (SSCCS-K-D; Bertrams, Unger, & Dickhäuser, 2011). Participants answered 10 items regarding how they felt that evening (e.g., 'I felt as if I do not have any willpower') using 4-point Likert-type scales (1 - not at all to 4 - very much). The mean response in this questionnaire was used as a measure of state self-control strength on a given day.

Stress-level assessment. Lacking an adequate German questionnaire for the assessment of perceived daily stress levels among student populations, we developed a new assessment tool for the present study. For this purpose, we pre-selected 23 items from an established daily-hassle scale (Holm & Holroyd, 1992) and translated them into German. Each item of this scale describes a daily event (e.g., 'I did not sleep well') that has the potential to be stressful. For each event, it must be indicated whether it happened within the last 24 h and, if so, how stressful it was on a 7-point Likert-type scale (1 - event happened but did not bother me at all to 7 - event happened and was unbearable). The pre-selected items were items we and a student assistant¹ considered as adequate for a student population (i.e., items describing events that could eventually occur to a student). Based on a pre-test ($N = 70$), we further selected those 14 items from the pre-selection that received a response value of one or higher (i.e., indication that this particular event happened to them) from more than 40% of the sample to ensure adequacy for the population of interest. This 14-item version yielded a good reliability in the pre-test ($\alpha = 0.80$). Mean response scores to these items were thus used for the assessment of perceived daily stress levels in the present study (see Table S2 in the

Supplemental Material, for a list of all items).

Self-efficacy scale. We assessed participants' perceived self-efficacy in the sport domain with the self-efficacy scale for physical exercise (SSA; Fuchs & Schwarzer, 1994). Participants have to indicate their perceived ability to maintain an exercise program on 12 items (e.g., 'I am confident that I can perform my planned physical activity even when I am tired') using 5-point Likert-type scales (1 - not confident at all to 5 - very confident). The mean response to all items was used as a measure of perceived self-efficacy.

HEXACO personality inventory. We used the revised 60-item version of the HEXACO personality inventory (HEXACO-PI-R; Ashton & Lee, 2009) to assess the personality dimensions Openness to Experience, Conscientiousness, Agreeableness, Extraversion, Emotionality, and Honesty/Humility. For each dimension, the mean response to the corresponding items was used as indicator. Personality dimensions were assessed because conscientiousness has been found to predict the exercise-intention-behaviour relationship in a previous study (Conner et al., 2007). However, to preview the present results, none of the personality dimensions predicted execution of the workout intention in the present study.

3. Results

We used a multilevel-modelling approach that allows for simultaneous consideration of state (i.e., Level 1) and trait (i.e., Level 2) predictors on workout execution. This approach accommodates for incomplete data sets. That is, the available data from participants who did not fill in the daily questionnaire on some day(s) was included in the analyses. With less than 10% missing values, the data quality of the present study was good. To our best knowledge, this is the first study investigating state-trait self-control strength effects in a population whose activity level is too low to gain health benefits from it, and thus we had no a priori assumptions about the parameter means and standard errors in this population. However, simulations studies by Maas and Hox (2005; see also Paccagnella, 2011) suggest that multilevel parameter estimates are of acceptable accuracy with at least 50 level-2 and 5 level-1 observations and our study ($N_{\text{level-2}} = 62$; $n_{\text{level-1}} = 7$) fulfilled these requirements.

3.1. Data preparation and descriptive results

Level-1 predictors were centred individually and Level-2 pre-



Fig. 2. Example of an exercise depicted in the workout manual.

¹ We are grateful to Lisa Scheffczyk for her support in the development of the stress assessment.

dictors were grand-mean centred to facilitate interpretation of beta estimates. Means, standard deviations, and reliability coefficients of all Level-2 predictors are presented in Table 2.

Workout-execution self-reports were used as the outcome

Table 2
Means, standard deviations, and reliability coefficients for all Level-2 predictors.

Measures	<i>M</i>	<i>SD</i>	Cronbach's α
Trait self-control strength (<i>N</i> = 61)	3.39	0.60	0.77
Social desirability scale (<i>N</i> = 61)	10.34	3.09	–
Self-efficacy scale (<i>N</i> = 61)	4.10	0.73	0.69
Hexaco personality dimensions (<i>N</i> = 59)			
Openness for experience	3.71	0.49	0.60
Agreeableness	3.25	0.52	0.66
Conscientiousness	3.52	0.62	0.80
Extraversion	3.44	0.57	0.70
Emotionality	3.40	0.63	0.77
Honesty/Humility	3.44	0.62	0.72

Note. Variations in sample size (*N*) are due to the fact that one participant did not fill in any questionnaire and two additional participants did not fill in the HEXACO-PI-R.

variable. In total, when considering each workout day of each participant as units of observation, there were 62 × 7 = 434 observations. Only infrequently the questionnaire was not filled in (i.e., 2.5% of all observations). On most days where the questionnaire was filled in, either all or none of the exercises were executed (87% of all valid observations). On only 1% of all valid observations less than half of the exercises were executed and on 12% of all valid observations more than half of the exercises were executed. As most people either performed all exercises or none on most days, workout-performance sum-scores on all seven days significantly deviated from a normal distribution according to Kolmogorov-Smirnov tests, *D*s > 0.36, *p*s < 0.001. We therefore recoded it into a binary format. Days on which participants performed equal to or less than half of the exercises were coded as zero, the other days were coded as zero. We applied this specific coding scheme because participants usually reported good reasons why they left out a few exercises (e.g., it was not possible to do it due to my leg pain), whereas they mostly mentioned motivational reasons when they performed only few exercises.² According to the (binary) workout-execution variable, the overall exercise adherence rate was quite good (*M* = 0.88; *SD* = 0.33) and individual exercise adherence rates ranged from 0.14 to 1.00.

3.2. Logistic multilevel analysis

Model analyses were conducted with the software *R* using the *lme4* library (Bates, Maechler, Bolker, & Walker, 2014). A logit-link function was used to relate the binary outcome variable to the continuous predictors. We first ran an exploratory full model containing all predictor variables (but no interactions) to identify significant predictors of workout likelihood. We then specified theory-based models to further investigate the research questions of interest, that is, the relative contributions of state and trait self-control strength to workout likelihood and the mediating role of state self-control strength for the relation between perceived daily stress and workout likelihood.

Full model. Perceived daily stress and state self-control strength were considered as Level-1 predictors; trait self-control strength, gender, HEXACO-PI-R factors, self-efficacy, and social desirability were considered as Level-2 predictors for workout likelihood. As evident from the modelling results displayed in Table 3, only state self-control strength was a significant predictor of the workout likelihood. Therefore, we did not include personality dimensions,

² Applying a more lenient coding scheme of assigning “1” to all observations, where at least one exercise was executed would not have changed the present results in any way. The basic pattern of results would have also be the same when applying a stricter coding scheme of assigning “1” to only those observations, where all exercises were executed.

Table 3
Multilevel modelling results of the full model and the state–trait model.

Predictors	<i>b</i>	<i>SE</i>	<i>z</i>	<i>p</i>
Full Model (<i>N</i> = 62)				
State self-control strength (Level 1)	2.39	0.81	2.96	0.003**
Daily stress (Level 1)	−0.72	0.77	−0.94	0.347
Trait self-control strength (Level 2)	−0.49	1.28	−0.38	0.705
Gender (Level 2)	1.21	1.65	0.73	0.465
Social desirability (Level 2)	0.12	0.22	0.55	0.579
Self-efficacy (Level 2)	0.77	0.69	1.12	0.263
Openness for experience (Level 2)	2.61	1.41	1.85	0.064
Agreeableness (Level 2)	−0.76	1.17	−0.65	0.515
Conscientiousness (Level 2)	−0.96	1.18	−0.81	0.419
Extraversion (Level 2)	−0.13	1.09	−0.12	0.903
Emotionality (Level 2)	−0.40	0.93	−0.43	0.668
Honesty/Humility (Level 2)	−0.72	1.03	−0.70	0.483
State-trait model of self-control strength (<i>N</i> = 62)				
State self-control strength (Level 1)	2.64	1.28	2.06	0.039*
Trait self-control strength (Level 2)	0.03	0.99	0.03	0.973
State × trait self-control strength	−0.12	0.70	−0.18	0.861

Note. *N* indicates sample sizes. **p* < 0.05; ***p* < 0.01.

self-efficacy, and social desirability as predictors in the following models.

Model of state–trait self-control strength. To further investigate the predictive power of state and trait self-control strength for workout likelihood, we specified a model with state self-control strength as Level-1 predictor, trait self-control strength as Level-2 predictor, and the cross-level interaction between the two self-control strength variables as additional predictor. Results are displayed in Table 3. State self-control strength was again a significant predictor of the likelihood to work out. State self-control and the cross-level interaction did not predict workout likelihood. Thus there was no evidence that trait self-control strength predicted workout likelihood over-and-above state self-control strength. Furthermore, there was no evidence that the predictive power of state self-control strength for workout likelihood depended on the trait self-control strength of a person.

Mediation model. In the full model perceived daily stress was not significant, implying that perceived stress levels were of minor importance for workout likelihood. However, when all other predictors were dismissed from the model, daily stress turned out to be a significant predictor of workout likelihood, *b* = −0.93, *SE* = 0.38, *z* = −2.47, *p* = 0.013. This finding indicates that the non-significant finding regarding daily stress in the full model could be due to the fact that the influence of daily stress on workout likelihood was mediated by the significant predictor state self-control strength. To test this hypothesis, which is illustrated in Fig. 1, we conducted a mediation analysis. Preacher, Zyphur, and Zhang (2010) suggest a multilevel mediation-test approach that differentiates between Level-1 and Level-2 predictors and provides a Sobel-like significance test for indirect effects, which seemed most appropriate for the present research question. Therefore, we used Preacher et al.'s *Mplus*-script for mediation tests in traditional multilevel models with predictor, mediator, and outcome variables assessed at Level 1, to test whether the relation between perceived daily stress and workout likelihood was mediated by state self-control strength. The script was adapted to accommodate for binary outcome variables. Results showed that there were significant effects of both daily stress on state self-control strength (Fig. 1, path a), *b* = −0.60, *SE* = 0.20, *z* = −3.05, *p* = 0.002, and of state self-control strength on workout likelihood (path b), *b* = 2.51, *SE* = 0.68, *z* = 3.68, *p* < 0.001. As described above, when daily stress was considered as the sole predictor for workout it had a significant effect, *b* = −0.93, *SE* = 0.38, *z* = −2.47, *p* = 0.013 (path c). The effect of daily stress on workout likelihood after controlling for state self-

control strength was not significant (path c'), $b = -0.33$, $SE = 0.51$, $z = -0.64$, $p = 0.525$. However, the indirect effect ($a \times b + cov_{ab}$) from daily stress to workout likelihood via state self-control strength was significant, $b = -1.36$, $SE = 0.59$, $z = -2.31$, $p = 0.021$. This pattern of results is completely in line with our hypothesis that perceived daily stress negatively affects the likelihood to work out but that this relation is mediated by the stress-induced reduction in state self-control strength.

4. Discussion

Regular physical activity has positive effects on physical and mental health (e.g., Bhui & Fletcher, 2000; Biddle & Mutrie, 2007; Goodwin, 2003; Ströhle, 2009). Although many people want to work out regularly, they may often fail to do so (e.g., Hallal et al., 2012). In the present study, we adopt the Strength Model of Self-Control (e.g., Baumeister et al., 1998) as a theoretical framework to the explanation of lapses in regular exercise execution among a population of healthy people that currently do not maintain an activity level that is known to foster health benefits but are motivated to increase their activity level. In doing so, we hope to account for some shortcomings of previous models (cf., e.g., Sniehotta et al., 2014; Rebar et al., 2015), that is, difficulties to explain why exercise execution is often not maintained over even short time periods. We started from the everyday observation that the likelihood of working out, after a (stressful) day requires a great deal of self-control strength, as one must inhibit the urge to relax and force oneself to perform rather straining physical exercises (cf., Ntoumanis, 2014). To test this assumption, we assessed state self-control strength as well as inter-individual differences in trait self-control strength and considered them as predictors for workout execution. In line with previous research, we found that self-control strength was lower on days that were perceived as stressful as compared to days that were not (Tangney et al., 2004). Additionally, both perceived stress level and state self-control strength were predictors of workout execution. Trait self-control strength, however, did not predict workout execution over-and-above state self-control strength. This latter finding was somewhat unexpected given that Bertrams and Englert (2013) found trait self-control strength to be predictive of workout execution in a student sample. However, the present multilevel approach and sample sizes were also better suitable for detecting level-1 predictors than level-2 predictor effects (cf., Maas & Hox, 2005). Further, we investigate individuals without established workout routines whereas participants in Bertrams and Englert's study already exercised regularly (i.e., on two days a week on average) and intended to increase their exercise frequency. Mean trait self-control strength and variances were comparable between the two studies, but it might well be that the role of trait self-control strength for workout behaviour depends on the initial activity level. In general, proximal predictors (i.e., state self-control) seem to be better predictors of actual behaviour than distal predictors (i.e., trait self-control; e.g., Flay, Snyder, & Petraitis, 2009). Further research is necessary to investigate this question.

From the present findings we conclude that—at least for individuals without established workout routines—states of self-control depletion are a critical caveat for regular workout execution. The present findings are an important step towards a better understanding of why people have particular problems with maintaining their workout plans over time. Further, the present findings are promising in as far as personality variables in general and trait self-control strength in particular seem to be less important than state self-control strength for the successful workout execution and the latter can be optimized much easier (e.g., Flay et al., 2009).

In this regard, the finding that the relation between perceived daily stress levels and workout execution was mediated by states of self-control strength is an important step in the development of interventions to improve workout maintenance. If stressful encounters indeed temporarily lower self-control resources that are required to execute workouts, any reduction of the daily stress level should benefit the workout likelihood.

A potential caveat to the present findings is that the assessment of state self-control was delayed until after the time point where participants should have performed their workout. We did this to not additionally remind or motivate participants to do the exercises with the questionnaire. However, the retrospective assessment of self-control may have caused participants to conclude that their self-control must have been low on days on which they did not exercise. We cannot completely rule out that such a retrospective bias was present in our data. However, most items of the questionnaire used to assess states of self-control strength are concerned with mental rather than physical states of fatigue (cf. Bertrams et al., 2011), which renders it unlikely that participants strongly relied on their exercising experience when answering them. Furthermore, exercise adherence only explained 37% of the variance in state self-control strength. Thus, despite the retrospective assessment, it was not the case that participants only took into account whether they exercised or not when judging their state of self-control strength.

Another potential caveat to the present findings is that they strongly rely on self-reports and could thus be affected by social-desirability effects. Because participants were instructed to execute their workouts on a daily basis, one could expect them to be dishonest regarding their actual workout behaviour. However, inter-individual differences in social-desirability traits did not affect workout likelihood. Therefore, it seems unlikely that social-desirability effects contaminated the present results.

Finally, from the observed correlational patterns we cannot be completely sure that the causal pathway was from perceived stress via state self-control strength to workout execution. Importantly, current theorizing in the area of self-regulation research suggests that dealing with stressful events taxes self-control resources that are then no longer available for subsequent acts of self-control (e.g., Tangney et al., 2004). Our interpretation is very much in line with this theorizing and is also supported by the mediation analysis results implying that the causal pathway is that daily stress affects self-control and self-control strength, in turn, affects workout adherence.

We do not consider the application of the Strength Model of Self-Control (e.g., Baumeister et al., 1998) as a completely new approach to explain workout execution but rather an extension of previous approaches (e.g., Hagger, 2010; Hagger et al., 2009). Indeed, there are theoretical overlaps between this model and other models of physical activity like the TPB (e.g., Ajzen, 1991). For instance, Schwarzer (1992) as well as Gollwitzer (1999) argue that volitional processes can support regular workout execution (see also Schwarzer & Luszczynska, 2008). This strategy seems to be effective also in health contexts (Hagger & Luszczynska, 2014). Recently, Achtziger, Gollwitzer, and Sheeran (2008) suggested using implementation intentions as a cognitive strategy to help individuals to follow their workout plans. This strategy requires individuals to generate specific intentions and to explicitly state when to execute them (e.g., “When I am coming home, then I will go for a run” rather than “I will do more sports”). Interestingly, implementation intentions have also been found to reduce the negative effects of ego depletion on subsequent self-control acts (e.g., Webb & Sheeran, 2009), suggesting that volitional processing helps dealing with reduced self-control strength.

We would also like to mention that the assumptions of the

Strength Model of Self-Control (e.g., Baumeister et al., 1998) have recently been challenged by several researchers, arguing that performance impairments from a first self-control task in a subsequent self-control task are not caused by a temporary depletion of a metaphorical strength. Instead it is argued that self-control impairments may be caused by motivational shifts (e.g., Inzlicht & Schmeichel, 2012), resource allocation (Beedie & Lane, 2012), or implicit theories about willpower (Job, Dweck, & Walton, 2010). Although the mechanism behind the ego depletion effect is not fully clarified yet, it is important to note that these alternative theoretical approaches do not neglect the existence of a self-control depletion effect, which is further supported by the strong effect sizes in a recent meta-analysis (Hagger et al., 2010). Although the process driving this effect is not of central significance for the present study, it is important to further investigate the causal mechanisms behind the self-control depletion effect. Whatever process it may be, however, the present research clearly demonstrates that, from an applied perspective, it is important to avoid states of self-control depletion as it may hinder regular workout execution.

In this regard, it seems promising that recent studies suggest that the ability to maintain self-control strength can be improved via training. For instance, regular self-control exertion over a two-week period can lead to better self-control in the long-run (e.g., Baumeister, Gailliot, DeWall, & Oaten, 2006). Furthermore, relaxation after a straining self-control task leads to a quicker revitalization of depleted self-control resources (Tyler & Burns, 2008). As far as we know, self-control trainings have not yet been adopted to explain regular exercise behaviour. Future studies should thus test whether self-control trainings have positive effects on the achievements of workout goals.

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Conflicts of interest

The authors hereby declare that there is no conflict of interest.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.psychsport.2016.06.001>.

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